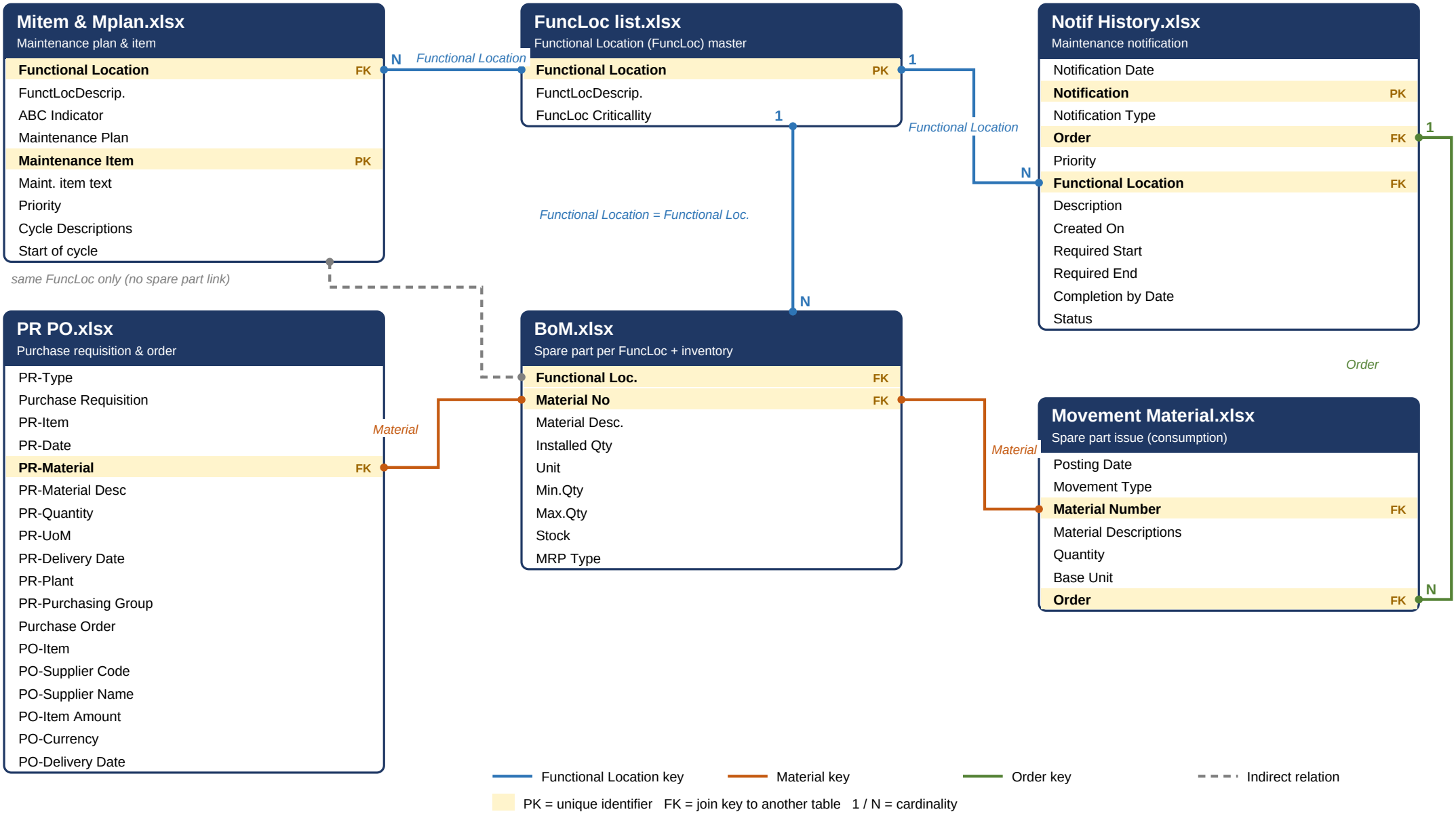
Case 3 - Spare Part Intelligence: Data Relationship Diagram

Caliber 2026 Hackathon Day | Baseline dataset (dummy) | 6 Excel files, 1 sheet "Data" each



Join Keys & Table Grain

#	From (parent)	To (child)	Join condition	Cardinality	Meaning
1	FuncLoc list	BoM	[Functional Location] = [Functional Loc.]	1 : N	Spare parts installed on / assigned to a FuncLoc
2	FuncLoc list	Notif History	[Functional Location] = [Functional Location]	1 : N	Failure / maintenance requests raised on a FuncLoc
3	FuncLoc list	Mitem & Mplan	[Functional Location] = [Functional Location]	1 : N	Preventive / planned maintenance scheduled on a FuncLoc
4	BoM	Movement Material	[Material No] = [Material Number]	-	Historical consumption of a spare part
5	BoM	PR PO	[Material No] = [PR-Material]	-	Procurement history of a spare part: date, quantity, supplier, price, delivery
6	Notif History	Movement Material	[Order] = [Order]	1 : N	Parts issued for the maintenance order (MO) of a notification
7	Mitem & Mplan	BoM	via [Functional Location]	indirect	Both tables refer to the same FuncLoc. Mplan & Mitem do not include the spare parts required to perform the PM work

Table summary

File	Casebook data source	One row represents	Main keys	Time column
FuncLoc list.xlsx	4. Equipment Criticality & Failure History	One functional location (FuncLoc / sub-FuncLoc / system)	Functional Location	-
BoM.xlsx	1. Spare Part Master Data; 3. Current Inventory Position	One material assigned to one FuncLoc	Functional Loc. + Material No	-
Movement Material.xlsx	2. Spare Part Consumption History	One material issue transaction	Material Number, Order	Posting Date
Notif History.xlsx	4. Equipment Criticality & Failure History	One maintenance notification	Notification; Order, Functional Location	Notification Date
PR PO.xlsx	5. Procurement & Lead-Time Data	One PR item linked to one PO item	Purchase Requisition + PR-Item; PR-Material	PR-Date, PO-Delivery Date
Mitem & Mplan.xlsx	6. Maintenance Plan	One maintenance item within a maintenance plan	Maintenance Item; Functional Location	Start of cycle

All data is synthetic (dummy) and generated for the Caliber 2026 Hackathon. It mirrors the structure and relationships of the original system data, not its actual values.